

DATA MANAGEMENT TECHNOLOGIES

UNIT 1 – TASK 1

TASK

Write down one for each of 1-5 in the case of a) only one entity, and b) several entities.

For each, explain briefly what is the object of interest (its name), what its properties are, and what its data are. Imagine adding more data and more properties to it. Then, write down how the database perspective simplify the storage and retrieval of data for each of these documents.

1. A text file
2. A word-processed file
3. An image
4. A Comma Separated Value file
5. An Excel file

What to submit, and how?

You will produce a mini-portfolio document that you will submit below. This should contain your answers for sub-tasks 1-5, and you will be receiving feedback for this task.

RESPONSE

1. Text file

a.) One entity:

- **Object:** Text document (e.g. .txt, .md file).
- **Properties:** filename, author, size, date created.
- **Data:** basic string content.
- **More data and properties:** version, tags, annotations and sections.

b.) Several entities: Each paragraph or each sentence within the text file can be treated as a single entity.

- **Object of interest:** Document, Paragraph, Sentence.
- **Properties:**
Document: title, author, date.
Paragraph: paragraph number.
Sentence: sentence number.
- **Data:**
Document: overall content.

Paragraph: paragraph text.

Sentence: sentence text.

More data and properties: text formatting (bold, italics, underlined).

Database perspective: Text can be separated into paragraphs then sentences.

The database could store three tables:

- **Document:** Includes a single record with information on the whole file.
- **Paragraphs:** Includes a record for each paragraph, including a paragraph-ID field.
- **Sentences:** Includes a record for each sentence, including a primary key called sentence-ID, and another key called sentence-text to store the text of the sentence.

2. Word-processed file

a.) One entity:

- **Object:** word-processed document (e.g. .doc or .docx file).
- **Properties:** filename, author, size, date created.
- **Data:** string content, images, charts, and graphs.
- **More data and properties:** language, version, tags, multiple versions, annotations, sections, and comments.

b.) Several entities: If the word document is treated as multiple entities, the file can be considered as multiple paragraphs or sentences, like the text file described above.

However, unlike a text file, a word document can also include tables, charts, and other diagrams.

These are analysed below.

- **Object of interest:** Tables, charts.
- **Properties:**
Tables: title, caption, rows, bars, and columns.
- **Data:**
Rows: basic string content.
Columns: basic string content.
Bars (bar charts, histograms): numeric content.

Database perspective: A database will simplify storing data in a word document. The word document text can be stored directly in the database. Alternatively, there can be hyperlink text stored in the database linking to specific sections in the document (such as paragraphs and diagrams).

3. Image

a.) One entity:

- **Object:** image or picture (e.g. .jpg, .jpeg, .png).
- **Properties:** name, dimensions, size, created date, resolution, bit depth, colour depth.
- **Data:** pixel array (RGB values per pixel).
- **More data and properties:** image label.

b.) Several entities:

- **Object:** image-tag, pixels.
- **Properties**
 - Tag: text.
 - Pixel: location (X-coordinate, Y-coordinate), colour.
- **Data:**
 - Tag: tag description.
 - Pixel: RGB value.
- **More data and properties:** image label.

Database perspective: The database can store images directly in a specialised field as a BLOB (Binary Large Object) file type.

However, it is common to store images outside the database, and store each image's related information, including file path, as a record in a table. As such, the user can search for the image through related information.

4. CSV file

a.) One entity:

- **Object:** Comma Separated Value file.
- **Properties:** file name, delimiter (comma), header row, number of rows and columns.
- **Data:** values arranged in rows and separated by commas.
- **More properties and data:** General descriptive information about the file, which can be stored outside the csv file.

b.) Several entities:

- **Object:** tables, rows and columns within the CSV file.
- **Properties:**
Column: column name, data type.
Row: row ID.
- **Data:** cell values.
- **More properties and data:** General descriptive data of the csv file.

Database perspective: A CSV file can be a useful replacement for a spreadsheet, as it is generally more lightweight.

5. Excel File

a.) One entity:

- **Object:** Excel file (.xlsx).
- **Properties:** file name, date created, author, number of sheets.
- **Data:** values arranged in rows and columns.
- **More properties and data:** diagrams, images, formulas, macros, version history.

b.) Several entities:

- **Object:** individual sheets or cells within the spreadsheet.
- **Properties:**
Individual sheet: sheet name, tab colour.
Cells: cell location, cell formatting (e.g. background and text colour).
- **Data:** cell values in the form of strings, floats, integers, etc.

Database perspective:

Each worksheet could be stored in the database as a table, the cells as records. The attributes could include cell font, background colour, etc.